

# Yuhang Zhang

+1 (979) 326-2667 | [yuhang.zhang@tamu.edu](mailto:yuhang.zhang@tamu.edu) | [linkedin.com/in/yuhangzhang3](https://www.linkedin.com/in/yuhangzhang3) | [yuhang-zhang17.github.io](https://yuhang-zhang17.github.io)

## EDUCATION

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### Texas A&M University

College Station, TX

*Ph.D. Candidate, Mechanical Engineering; GPA: 4.0/4.0*

*Jan 2023 – May 2027 (Expected)*

- Dissertation: *Modeling and Optimal Control of District Heating and Cooling Systems*
- Advisor: Prof. Zheng O'Neill

### Tongji University

Shanghai, China

*Master of Engineering in Power Engineering; GPA: 4.9/5.0*

*Sep 2019 – Mar 2022*

- Thesis: *Study on Leakage Diagnosis Method for Air Conditioning Water System Networks*

### Tongji University

Shanghai, China

*Bachelor of Science in Building Environment and Energy Engineering; GPA: 4.6/5.0*

*Sep 2015 – Jul 2019*

## RESEARCH INTERESTS

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- Building/HVAC&R Energy Efficiency
- Intelligent Building Controls and Optimization
- District Heating and Cooling Systems (5GDHC)
- Heat Pump Technologies
- Model Predictive Control (MPC)
- Thermal Energy Storage (TES)
- Building Decarbonization and Electrification

## RESEARCH EXPERIENCE

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### NSF PIRE: Building Decarbonization via AI-empowered District Heat Pump Systems

*National Science Foundation (NSF)*

*Jan 2023 – Present*

- Developed a high-fidelity virtual testbed of district heat pump systems integrating physics-based Modelica thermal models with AI-driven predictive control frameworks.
- Designed Model Predictive Control (MPC)-based optimal control frameworks for intelligent district energy system operation.
- Applied the framework to assess renewable and waste heat integration (e.g., data center heat recovery), demonstrating improved efficiency and carbon reduction potential.

### DoD ESTCP: Building Energy Efficiency through Thermal Microgrids – Fort Hood Phase I Feasibility Study

*Department of Defense (DoD) ESTCP*

*Jan 2024 – Jan 2025*

- Developed and validated Modelica digital twins and machine-learning load forecasting models for a 30-building military installation district energy system.
- Implemented MPC optimization for chiller plant operation and temperature reset strategies.
- Demonstrated up to ~70% modeled energy reduction; performed 30-year life-cycle cost analysis.

### DOE: Solar-Geothermal District Heating and Cooling (Sol-GDHC) System – Citizen Potawatomi Nation

*Department of Energy (DOE)*

*Dec 2023 – Sep 2024*

- Contributed to the design of a 100% electrified geothermal district energy system for an Oklahoma tribal community.
- Developed an integrated simulation framework for system-level performance evaluation and annual operation analysis.
- Developed detailed Modelica component-level models of hybrid photovoltaic thermal (PVT) panels.
- Supported optimization of geothermal borehole design and loop temperature control strategies.
- Reduced projected electricity consumption by 38% and eliminated natural gas use in the retrofit scenario.

### Master's Thesis: Leakage Diagnosis in Air Conditioning Water System Networks

*Tongji University*

*Sep 2020 – Mar 2022*

- Modeled thermo-hydraulic dynamics of a complex piping network and validated results against field measurement data.
- Developed a parameter identification approach for pipeline hydraulic resistance using metaheuristic optimization.
- Proposed a 2-stage leakage fault detection and diagnostics method using Adam-optimized neural networks.

## PROFESSIONAL EXPERIENCE

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### PhD Research Intern

Jun 2025 – Aug 2025

*Pacific Northwest National Laboratory (PNNL)*

*Richland, WA*

- Validated DOE's TES Sizing Tool by executing 240+ parametric simulations across time-of-use and dynamic pricing tariff scenarios.
- Designed optimized Thermal Energy Storage (TES) dispatch control strategies, achieving up to 85% simulated electricity cost reduction.
- Collaborated with DOE's Stor4Build consortium to improve tool robustness for grid-interactive efficient buildings (GEBs).

### Graduate Research Assistant

Jan 2023 – Present

*Texas A&M University*

*College Station, TX*

- Develop high-fidelity simulations for building and district energy systems using Modelica/Dymola and Python.
- Conduct research on predictive controls and optimization for energy-efficient smart energy systems.
- Contribute to NSF-, DoD-, and DOE-funded research projects on building decarbonization and district energy.

### HVAC Engineer

Jul 2022 – Dec 2022

*Envision Group*

*Shanghai, China*

- Led development of AHU/chiller control optimization and fault detection & diagnostics (FDD) features for a smart building energy management platform.
- Coordinated with engineering and sales teams to align product design with client needs and deployment requirements.

## PUBLICATIONS

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### Journal Articles

- [1] Liu, M., Zhang, Y., Yang, Z., & O'Neill, Z. (2026). Comparative Study of Energy Flexibility in Model Predictive Control of Single- and Variable-Speed Heat Pumps with Active Thermal Energy Storage. *Energy*, 141387. doi:10.1016/j.energy.2026.141387
- [2] Liu, M., Zhang, Y., Yang, Z., & O'Neill, Z. (2026). A nationwide assessment of energy flexibility from predictive-controlled heat pump and active thermal energy storage system for building electrification. *Building Simulation*. doi:10.1007/s12273-026-1421-4
- [3] Zhang, Y., Liu, M., O'Neill, Z., & Wen, J. (2024). Temperature control strategies for fifth-generation district heating and cooling systems: A review and case study. *Applied Energy*, 376, 124156. doi:10.1016/j.apenergy.2024.124156
- [4] Liu, R., Zhang, Y., & Li, Z. (2022). Leakage Diagnosis of Air Conditioning Water System Networks Based on an Improved BP Neural Network Algorithm. *Buildings*, 12(5). doi:10.3390/buildings12050610

### Conference Proceedings

- [1] Jiang, Z., Song, R., Li, G., Zhang, Y., O'Neill, Z., Wang, X., Goldfeder, J., & Dong, B. (2026). BESTOpt: A Modular, Physics-Informed Machine Learning Based Building Modeling, Control and Optimization Framework. *SimBuild 2026* (accepted).
- [2] Zhang, Y., Liu, M., O'Neill, Z., & Wen, J. (2025). Operating temperature optimization for a 5th-generation district heating and cooling network using model predictive control. *2025 ASHRAE Winter Conference*, Orlando, FL. doi:10.63044/w25zha66
- [3] Li, G., Zhang, Y., Liu, M., O'Neill, Z., Song, L., Wang, G., Cai, J., & Wu, X. (2025). Modeling and integration of photovoltaic thermal (PVT) hybrid solar collectors in a fifth-generation district heating and cooling network. *Building Simulation 2025*. doi:10.26868/25222708.2025.1393
- [4] Zhang, Y., Liu, M., Yang, Z., Calfa, C., & O'Neill, Z. (2024). Development and validation of a water-to-air heat pump model using Modelica. *American Modelica Conference 2024*, 119–126. **Best Student Paper**. doi:10.3384/ECP207119
- [5] Zhang, Y., Liu, M., O'Neill, Z., & Wen, J. (2024). A comparative analysis of energy performance of ultra-low temperature district heating and cooling networks with different substation configurations. *ASHRAE Transactions*, 130, 796–805. doi:10.63044/w24zha94
- [6] Guan, H., Zhang, Y., O'Neill, Z., & Dong, B. (2026). A Scientific Machine Learning Approach for Geothermal Borefield Thermal Modeling via Residual Learning. *9th International High Performance Buildings Conference at Purdue*, West Lafayette, IN.

### Under Review

- [1] Guan, H., Zhang, Y., O'Neill, Z., & Dong, B. (2026). Adaptive Fusion: A Scientific Machine Learning Framework for Geothermal Borefield Thermal Modeling. *Under review at Energy*.

### In Preparation

- [1] Zhang, Y., Liu, M., Wen, J., & O'Neill, Z. (2026). A High-Fidelity Model Predictive Control Framework for Temperature Optimization in 5th-Generation District Heating and Cooling Networks. *Manuscript in preparation*.
- [2] Zhang, Y., Guan, H., Hankinson, N., & O'Neill, Z. (2026). Dynamic Techno-Economic Assessment of Data Center Waste Heat Recovery for Low-Temperature District Heating under Varying Waste Heat Availability and Temperature. *Manuscript in preparation*.
- [3] Zhang, Y., Guan, H., O'Neill, Z., & Wen, J. (2026). A Physics-Based Framework for System-Level Fault Impact Analysis in Fifth-Generation District Heating and Cooling Systems. *Manuscript in preparation*.

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## CONFERENCE PRESENTATIONS (WITHOUT PROCEEDINGS)

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- “Feasibility of Geothermal Borefield Integration in a 5th-Generation District Thermal Energy Network: A Case Study of an Oklahoma Tribal Community” — *Texas A&M Geothermal Research Opportunities Symposium*, Apr 2025. **2nd Place, Poster Competition**.
- “Building Decarbonization with a 5th Generation District Heating and Cooling (5GDHC) System: A Case Study of an Oklahoma Tribal Community” — *Texas A&M Conference on Energy, College Station, TX*, Sep 2024.
- “A Comparative Analysis of Energy Performance of Ultra-low Temperature District Heating and Cooling Networks with Different Substation Configurations” — *BAS2024 – Building Automation Symposium, Syracuse, NY*, May 2024.

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## ACADEMIC SERVICE

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### Peer Review

- Reviewer, *Science and Technology for the Built Environment* (2025–Present)
- Reviewer, ASHRAE Conferences – Winter & Annual (2025–2026)
- Reviewer, AEI International Research & Design Conference (2026)

### ASHRAE Technical Committees

- Corresponding Member, TC 7.5 – Smart Building Systems (2025–Present)
- Provisional Corresponding Member, TC 6.2 – District Energy (2023–Present)
- Provisional Corresponding Member, TC 6.9 – Thermal Storage (2023–Present)
- Provisional Corresponding Member, TC 6.8 – Geothermal Heat Pump Systems (2023–Present)

### Leadership

- ASHRAE Student Branch Officer, Texas A&M University (2023–2024): organized monthly technical seminars and promoted student engagement in ASHRAE activities and conferences.

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## HONORS & AWARDS

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- ASHRAE Graduate Student Grant-in-Aid Award (2026)
- J. George H. Thompson Graduate Fellowship (2025)
- Mechanical Engineering Graduate Summer Research Grant (2024–2025, two awards)
- Mechanical Engineering Graduate Student Travel Award (2023–2025, three awards)
- Best Student Paper, American Modelica Conference 2024
- 2nd Place Poster, Texas A&M Geothermal Research Symposium (2025)
- Tongji University Scholarships of Excellence (2015–2018, four awards)
- National Endeavor Scholarship (2015–2016)

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## TECHNICAL SKILLS

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**Programming & Data:** Python, MATLAB, C/C++

**Modeling & Simulation:** Modelica/Dymola, EnergyPlus, Ansys Fluent, EES

**Controls & Optimization:** Model Predictive Control (MPC), Metaheuristic Optimization, Machine Learning (Neural Networks, Load Forecasting)

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## REFERENCES

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**Zheng O'Neill**, Professor, Mechanical Engineering; J. Mike Walker '66 Professor II, Texas A&M University  
[zoneill@tamu.edu](mailto:zoneill@tamu.edu) | +1 (979) 845-4931